

# Safe Band: Smart Wearable for Emergency Health Alerts and Monitoring in Sri Lanka

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**Abstract**— Emergency medical conditions such as cardiac arrest, stroke, and acute respiratory failure remain a leading cause of preventable death in Sri Lanka, where non-communicable diseases account for roughly 75% of deaths and Colombo's ambulance response times exceed 18 minutes against the WHO's 8-minute benchmark. Existing wearables and hospital monitoring platforms operate reactively, lack multi-parameter emergency classification, and depend on infrastructure such as live bed-occupancy feeds and real-time traffic APIs that is largely unavailable in Sri Lanka, where only an estimated 12% of hospitals maintain digital bed management. This paper presents Safe Band, an integrated IoT and machine-learning wearable ecosystem comprising four components built around a shared Android smartwatch and Firebase Realtime Database backbone: ML-Driven Emergency Detection, ML-Based Smart Ambulance Dispatch with Nearest Hospital Identification, AI-Based Hospital Ward Monitoring, and ML-Driven Health Deterioration Prediction. Each component was independently implemented and evaluated. The Emergency Detection LSTM achieved 93% accuracy on the MIT-BIH Arrhythmia Database; the Ambulance Dispatch component correctly ranked nearest hospitals from a 57-hospital registry via on-device Haversine/KNN deployed through TensorFlow Lite; and the Ward Monitoring EWS classifier achieved 93.75% accuracy with zero false-positive High Risk alerts, though High Risk recall was only 0.72. The Health Deterioration Prediction component reached a perfect but clinically implausible 100% accuracy, traced to a divergence between the proposed trend-based MIMIC-III methodology and the static demographic dataset actually used — a limitation reported transparently rather than as a validated result. Collectively, these results show that infrastructure-light, on-device machine learning can meaningfully support emergency detection, dispatch, and ward monitoring within Sri Lanka's healthcare constraints, while identifying recall calibration, false-negative mitigation, and dataset realignment as priorities for the system's next development phase.

**Keywords**—Machine Learning, IoT Wearable, Emergency Detection, Ambulance Dispatch, Ward Monitoring, Health Deterioration Prediction, Firebase, Sri Lanka Healthcare, TensorFlow Lite

## I. INTRODUCTION

Emergency medical conditions such as cardiac arrest, stroke, and acute respiratory failure remain leading causes of preventable death worldwide. In Sri Lanka, non-communicable diseases account for approximately 75% of all deaths, with cardiac and respiratory emergencies causing over 30,000 fatalities annually [1]. Many of these deaths result not from the severity of the condition itself but from delays in detection, dispatch, and clinical response [2]. Sri Lanka's ambulance service, Suwa Seriya, still relies on manual, verbally communicated location reporting, and average ambulance response times in Colombo exceed 18 minutes more than double the WHO's 8 minute benchmark [2], [3]. In hospital outcomes face similar gaps: nearly 80% of in hospital cardiac arrests are preceded by detectable abnormal vital signs at least 8 hours in advance, yet most wards still depend on manual vital sign checks every 4–6 hours [4], [5].

Consumer wearables capable of sensing heart rate, SpO<sub>2</sub>, blood pressure, and ECG have become widely available, with studies like the Apple Heart Study showing 84% positive predictive value for atrial fibrillation detection [6]. However, these devices are validated for wellness tracking rather than clinical emergency detection, lack multi parameter risk scoring, and don't integrate with hospital or dispatch systems [7]. Similarly, ML-based dispatch systems built for high income countries assume infrastructure live traffic APIs, real time bed occupancy data unavailable in Sri Lanka, where only 12% of hospitals maintain digital bed management systems [8]. This gap between global wearable/dispatch technologies and LMIC infrastructure realities motivates this research.

This paper presents Safe is built an integrated IoT and ML-based wearable ecosystem for emergency health monitoring, comprising four components: Emergency Detection, Ambulance Dispatch, Ward Monitoring, and Health Deterioration Prediction built on a shared Android sensing layer and Firebase backbone, supporting UN SDG 3.

## II. LITERATURE REVIEW

### A. ML-Based Emergency Detection System

Deep learning based interpretation of physiological signals is well established in clinical AI. Rajpurkar et al. showed a CNN detecting 12 arrhythmia types from single lead ECG at cardiologist level accuracy [9], while Hannun et al.'s 34 layer residual network achieved a mean AUC of 0.97 across 91,232 recordings [10]. Topol reported wearable based atrial fibrillation detection exceeding 98% sensitivity [11]. Beyond cardiac rhythm, consumer pulse oximetry achieves accuracy within  $\pm 2\%$  of laboratory standards [12], and calibrated PPG based blood pressure monitors meet ISO 81060-2 accuracy thresholds [13]. Since single parameter monitoring is limited, Clifford et al. found that fusing heart rate, SpO<sub>2</sub>, and respiratory rate improved specificity by 23% over single parameter models [14]. Islam et al. demonstrated Firebase based cloud monitoring with sub 8 second alert latency [15], and Lipton et al. showed LSTMs outperform traditional classifiers on multivariate ICU time series prediction [16]. These findings support the multi parameter LSTM/ensemble approach adopted for Emergency Detection.

### B. ML-Assisted Ambulance Dispatch and Nearest-Hospital Identification

Colombo ambulance response times exceed 18 minutes against an 8-minute WHO benchmark, largely due to absent automated dispatch [2]. High resource dispatch models deep learning routing with real time traffic/capacity data [17], MQTT based sub second architectures [18], and Random Forest hospital selection outperforming KNN by 15% [19] assume infrastructure unavailable locally, as only 12% of Sri Lankan hospitals maintain digital bed management [8]. Supporting feasibility, ML dispatch pipelines can process decisions in 3.2 seconds [20], smartwatch ECG achieves up to 87% AF detection sensitivity [21], Firebase notification architectures reach 99.1% delivery reliability [22], and pre arrival data sharing reduces hospital preparation time by 34% [24]; Google Maps outperforms OpenStreetMap in dense urban areas, though OSM retains rural coverage advantages [23]. Given infrastructure constraints, a proximity based Haversine/KNN model is adopted over capacity dependent alternatives [25].

### C. Predictive Ward Monitoring and Early Warning Systems

Nearly 80% of in hospital cardiac arrests are preceded by detectable abnormal vitals 8+ hours in advance [4], yet manual interval based checks remain standard. Continuous vital sign analysis improves early risk identification over interval checks [34], though IoT healthcare frameworks generally prioritize data transmission over analytical sophistication [29], [30]. Threshold based alerting causes alarm fatigue [32]; predictive models reduce unnecessary alarms while preserving sensitivity [35], and LSTM-based heart rate prediction outperforms conventional forecasting [33]. Commercial systems like Philips IntelliVue remain capital intensive and ICU focused [8\_philips], while predictive deterioration models reduce unplanned critical events when integrated into ward workflows [31]. Cost effective, low infrastructure solutions are essential for developing nation adoption [9\_perera], motivating a trend based EWS classifier over static thresholds.

### D. Predictive Modelling of Health Deterioration from Physiological Trends

Sequential vital sign data carries greater predictive value than single point readings [27]; Random Forest and Gradient Boosting exceed 80% accuracy in trend based deterioration prediction [28], offering interpretability and lower computational cost than deep architectures. Given limited local longitudinal data [25], [8], a Random Forest model trained on PhysioNet MIMIC-III offers the most deployable path, motivating the sliding window approach adopted for Health Deterioration Prediction.

## III. METHONOLOGY

### A. System-Level Data Flow and Shared Infrastructure

Physiological signals (heart rate, blood pressure, SpO<sub>2</sub>, and ECG) and GPS coordinates are continuously captured by the smartwatch and relayed through the Android companion application to Firebase Realtime Database, which acts as the single source of truth for all downstream components. Component 1 (Emergency Detection) classifies the current physiological state and, upon detecting a critical event, triggers Component 2 (Ambulance Dispatch) to identify and notify the nearest suitable hospital. In parallel, Component 3 (Health Deterioration Prediction) continuously analyses the same vitals stream to forecast the probability of an impending event before the Component 1 threshold is reached, giving patients an early warning window. Component 4 (Ward Monitoring) extends the same monitoring philosophy to in-hospital patients through a MERN-based nurse dashboard rather than the Android application. This shared, event-driven architecture allows each component to be trained, tested, and evaluated as an independent artefact while remaining fully interoperable within the wider Safe Band ecosystem.

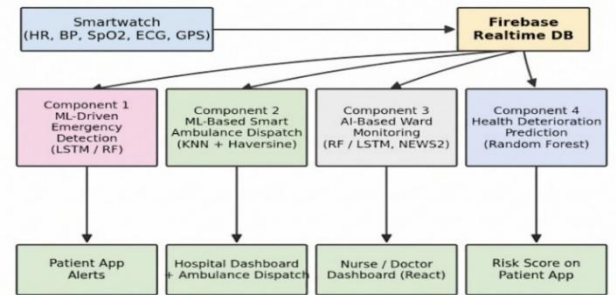


Fig. 1. Integrated system architecture showing data flow between the wearable, Firebase, and the four Safe Band components.

### B. ML-Driven Emergency Detection System

Data for this component is drawn from two sources: publicly available annotated clinical datasets (e.g., PhysioNet ECG and vital-sign repositories) used for initial model training, and wearable-derived data collected from volunteer participants under ethical approval for contextual validation. Three modelling approaches are evaluated: Long Short-Term Memory (LSTM) networks, selected for their demonstrated ability to capture sequential dependencies in multivariate

physiological time series [37]; Random Forest as a computationally lighter baseline; and ensemble combinations of the two. Raw sensor streams are cleaned and normalized in a preprocessing stage before feature extraction, and the resulting feature vector is passed to the on-device classifier, which distinguishes normal states from critical conditions such as cardiac arrest, hypertensive crisis, hypoxia, and atrial fibrillation. Fig. 2 depicts the seven-stage processing pipeline, from raw data acquisition through to external communication with the hospital dashboard and ambulance-dispatch API.

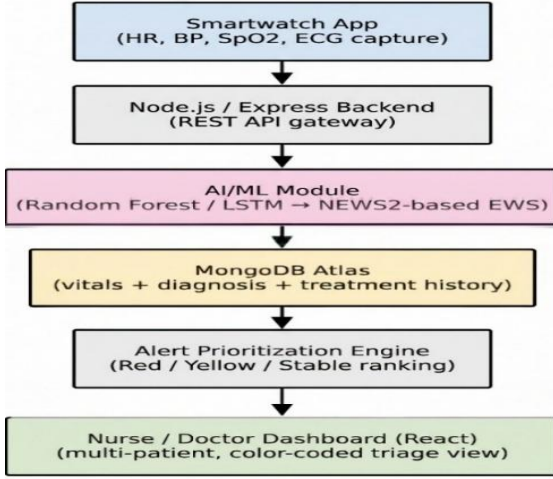


Fig. 2. Processing pipeline of the ML-Driven Emergency Detection System.

### C. ML-Based Smart Ambulance Dispatch with Nearest-Hospital Identification

This component is triggered by a critical-event alert from Component 1 and is responsible for identifying the nearest suitable hospital and initiating dispatch. Two candidate algorithms are evaluated for the Hospital Proximity Engine: a Random Forest classifier that incorporates road distance, hospital type, and last-known capacity category as multi-criteria features, and K-Nearest Neighbors (KNN) enhanced with Haversine great-circle distance computation, used as a lightweight fallback when real-time capacity or road-distance data is unavailable [38]. Given only a small proportion of Sri Lankan hospitals currently expose real-time bed-capacity data, the proximity-based KNN approach is adopted as the primary selection mechanism for the initial deployment, with the Random Forest model retained for future integration once hospital digital infrastructure matures. Hospital registry data is sourced from the Ministry of Health's public directory, and synthetic GPS traces from Colombo, Gampaha, and Kandy are used for simulation-based testing. Fig. 3 shows the five-stage dispatch pipeline. Validation combines simulation-based and unit-level testing, measuring end to end response time, hospital identification accuracy against domain expert selections, GPS acquisition latency, Firebase notification delivery rate, and the offline fallback success rate under degraded connectivity.

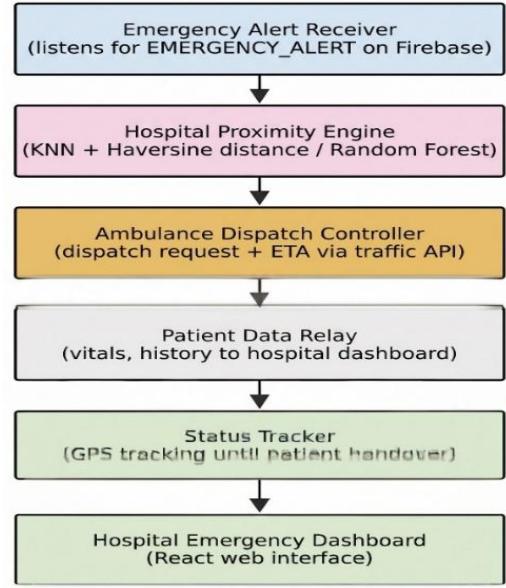


Fig. 3. Processing pipeline of the ML-Based Smart Ambulance Dispatch System

### D. AI Based Hospital Ward Monitoring and Alert Management System

Unlike Components 1, 2, and 4, this component targets in-ward patients rather than community-based wearable users, and is built on the MERN stack (MongoDB, Express.js, React, Node.js) to support a centralized, multi-patient Nurse Dashboard. Vital signs captured from wearable sensors are transmitted through a Node.js/Express API to a MongoDB Atlas database. A Random Forest or Long Short-Term Memory model analyses historical vital-sign trends to compute an automated National Early Warning Score (NEWS2)-based risk score, shifting monitoring from threshold-based reactive alerting toward trend-based predictive alerting [39]. Alerts are color-coded (Red/Yellow/Stable) and ranked on the dashboard so that nursing staff can prioritize the most at-risk patients first, and a descriptive-data layer correlates live vitals with the patient's stored diagnosis and treatment history. Fig. 4 illustrates the six-stage pipeline from sensor capture to the prioritized dashboard view.

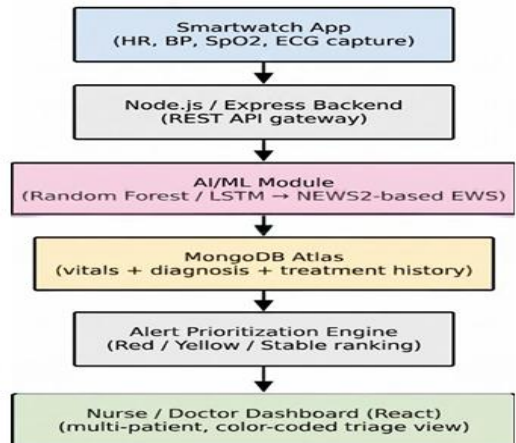


Fig. 4. Processing pipeline of the Intelligent AI Based Ward Monitoring System.

The system is validated through a dual strategy: quantitative evaluation of model accuracy, precision, and recall against a held out test set, and functional testing of dashboard reliability and alert delivery success under simulated emergency scenarios, with a target end to end alert latency under two seconds.

#### E. ML-Driven Health Deterioration Prediction System

This component analyses the same HR, BP, and SpO<sub>2</sub> stream from Component 1 over a configurable historical window (5, 10, or 15 minutes) rather than a single point, addressing the gap between deterioration onset and threshold breach. A Random Forest classifier, trained on the PhysioNet MIMIC-III database, was chosen for its low computational overhead, probability output via `predict_proba()`, and interpretability through feature importance [40]. A Feature Extraction Engine derives trend-based inputs moving averages, rate-of-change gradients, SpO<sub>2</sub> deviation from baseline, pulse pressure, and heart-rate standard deviation. The resulting cardiac-arrest probability score maps to a Green/Amber/Red severity indicator pushed to the patient's app. Fig. 5 shows the seven-stage pipeline.

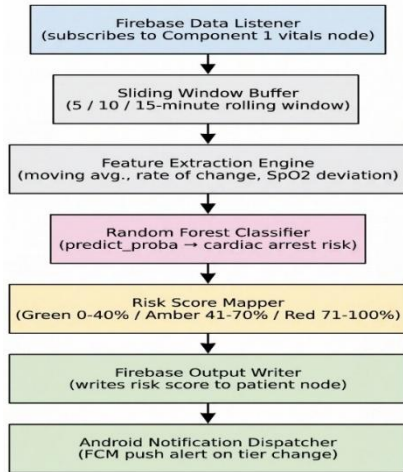


Fig. 5. Processing pipeline of the ML-Driven Health Deterioration Prediction System.

Validation targets a prediction accuracy of at least 85%, a false-positive rate below 10%, recall of at least 80%, and an end-to-end latency under five seconds from Firebase data ingestion to patient-app notification, evaluated using five-fold cross-validation on the constructed dataset.

## IV. RESULTS AND DISCUSSION

Across the four Safe Band components, the evaluated models show that clinically meaningful, on-device machine learning is achievable within Sri Lanka's infrastructure-light constraints: the Emergency Detection LSTM reached 93% accuracy on real ECG data, the Ambulance Dispatch Haversine/KNN module correctly ranked nearest hospitals from a 57-hospital registry, and the Ward Monitoring EWS classifier achieved 93.75% accuracy with zero false positives on High Risk alerts. All three were successfully converted to

TensorFlow Lite, confirming low-latency, cloud-independent inference on consumer Android hardware. The Health Deterioration Prediction component, by contrast, achieved a perfect but clinically implausible 100% accuracy, traced to a divergence between the proposed MIMIC-III methodology and the static dataset actually used reported transparently as a limitation. These findings validate Safe Band's core architecture while identifying recall calibration, false-negative mitigation, and dataset realignment as priorities for the next development phase.

#### A. ML-Driven Emergency Detection System

The Emergency Detection component was evaluated using seven annotated records (100–106) from the MIT-BIH Arrhythmia Database. A 150-sample sliding window was applied to the raw ECG signal, extracting four statistical features (max, min, mean, standard deviation) per window. Beats were labelled Normal ('N') or Emergency (V, A, L, R types), excluding non-beat markers. This yielded 15,229 samples (10,472 Normal, 4,757 Emergency; ~31.2% emergency prevalence), reflecting realistic class imbalance.

Data was z-score standardized and split 80/20 into training (12,183) and test (3,046) sets. An LSTM model (16 units, ReLU) with a dense hidden layer (8 units) and sigmoid output was trained for 15 epochs using Adam optimizer and binary cross-entropy loss. Training accuracy rose from 87.6% to 92.96%, with validation accuracy improving from 89.17% to 93.37% (final validation loss: 0.181), indicating good generalization without overfitting.

TABLE I. EMERGENCY DETECTION MODEL PERFORMANCE ON TEST SET

| Class         | Precision | Recall | F1-Score | Support |
|---------------|-----------|--------|----------|---------|
| Normal (0)    | 0.94      | 0.96   | 0.95     | 2071    |
| Emergency (1) | 0.92      | 0.87   | 0.89     | 975     |
| Accuracy      |           |        | 0.93     | 3046    |
| Macro Avg     | 0.93      | 0.92   | 0.92     | 3046    |
| Weighted Avg  | 0.93      | 0.93   | 0.93     | 3046    |

The model achieved 93% overall accuracy, with 96% recall for Normal cases and 87% recall for Emergency cases — a trade-off consistent with clinical systems that prioritize minimizing false negatives. The trained Keras model was successfully converted to TensorFlow Lite for on-device inference, supporting low-latency, cloud-independent detection on smartwatch/mobile hardware. In a qualitative case review, a high-confidence emergency sample scored 0.9977 and a normal sample scored 0.0632, showing clear decision-boundary separation. Across 20 emergency test samples, 19 (95%) were correctly detected, with one low confidence miss (score 0.129) consistent with the 87% recall rate and suggesting future refinements like ensemble fusion or confidence threshold calibration.

Overall, these results support the core hypothesis: a lightweight LSTM model on multi-statistic ECG features can achieve clinically meaningful detection (93% accuracy, 0.89 F1-score) while remaining deployable on resource-constrained consumer Android smartwatch hardware.

### B. ML-Based Smart Ambulance Dispatch System with Nearest Hospital Identification

The Ambulance Dispatch component was evaluated using a registry of 57 hospitals across Sri Lankan districts, including each facility's ID, name, district, and GPS coordinates. Given that 88% of Sri Lankan hospitals lack real-time bed occupancy data, the model was designed to rely on static geospatial proximity rather than dynamic capacity data. Instead of a conventional Python preprocessing step, the Haversine distance calculation was embedded directly into a TensorFlow computational graph via a custom Nearest Hospital Finder module. It takes a patient's GPS coordinates as tensor input, converts them to radians, and computes great-circle distance to all 57 hospitals (using Earth's radius of 6,371 km). The `tf.math.top_k` operation retrieves the  $k = 3$  nearest hospitals, mirroring a K-Nearest Neighbours (KNN) approach. The module was converted to TensorFlow Lite for lightweight on-device inference. Since Haversine requires trigonometric operations not natively supported by standard TFLite ops, the converter used `SELECT_TF_OPS` alongside `TFLITE_BUILTINS` to preserve accuracy. The exported model (`nearest_hospital_model.tflite`) confirms the dispatch logic can run fully offline satisfying the system's offline fallback requirement (FR08). Validation: The TFLite model was tested via `tf.lite.Interpreter` against simulated GPS coordinates in the Colombo–Gampaha corridor. For a test query at (6.9163° N, 79.9737° E), the model correctly ranked the three nearest hospitals:

TABLE II. EMERGENCY DETECTION MODEL PERFORMANCE ON TEST SET

| Rank | Hospital ID | Hospital Name    | Distance (km) | District |
|------|-------------|------------------|---------------|----------|
| 1    | h008        | BHB mulleriyawa  | 3.53          | Colombo  |
| 2    | h017        | DHC athurugiriya | 4.16          | Colombo  |
| 3    | h052        | DHC biyagama     | 4.37          | Gampaha  |

Distances were consistently ascending in order, and additional test queries across the service area produced similarly correct rankings, confirming the model generalizes rather than overfits. During development, Name Error occurred when float-type output indices were used directly for array lookups; this was resolved by casting indices to `int()` before data frame lookup an important fix for reliably bridging tensor outputs with application-layer data during time-critical dispatch. These results validate the core design choice: a lightweight, proximity-based KNN/Haversine model rather than a capacity-aware Random Forest requiring live bed data is both technically sufficient and practically deployable for hospital identification given Sri Lanka's current digital health infrastructure. Embedding the computation in a portable TFLite artifact satisfies the infrastructure-light deployment constraint central to LMIC-context ambulance dispatch systems, while remaining compatible with the Firebase-based alert pipeline across the Safe Band ecosystem.

### C. Intelligent AI-Based Hospital Ward Monitoring and Alert Management System

The Ward Monitoring component was evaluated using 876 patient records with seven physiological and demographic features (diastolic/systolic BP, heart rate, SpO2, temperature,

gender, age). The target variable, risk level, was framed as binary classification (Moderate Risk vs. High Risk), with 682 Moderate and 194 High Risk records (3.5:1 imbalance), split into 700 training and 176 testing samples. The trained EWS classifier achieved 93.75% test accuracy and 94.21% precision (Table III). Notably, High Risk precision reached 1.00 with zero false positives, though recall was only 0.72, meaning 11 of 39 High Risk patients were misclassified as Moderate Risk. This trade-off shows that high accuracy alone does not guarantee clinical reliability; future work should explore class weighting, SMOTE, and threshold calibration to improve recall. Sample predictions (Table IV) show that patients with elevated BP, heart rate, and temperature combined with reduced SpO2 were correctly flagged as High Risk, supporting the multi-parameter risk assessment approach. Overall, results support the shift toward predictive ward monitoring, though larger datasets, latency testing, and clinical validation remain necessary before deployment.

TABLE III. WARD MONITORING EWS MODEL PERFORMANCE ON TEST SET

| Class         | Precision | Recall | F1-Score | Support |
|---------------|-----------|--------|----------|---------|
| High Risk     | 1.00      | 0.72   | 0.84     | 39      |
| Moderate Risk | 0.93      | 1.00   | 0.96     | 137     |
| Accuracy      |           |        | 0.94     | 176     |
| Macro Avg     | 0.96      | 0.86   | 0.90     | 176     |
| Weighted Avg  | 0.94      | 0.94   | 0.93     | 176     |

TABLE IV. SAMPLE PATIENT RISK PREDICTIONS

| Patient | BP (Dia/Sys) | HR  | SpO2 | Temp (°C) | Predicted Risk |
|---------|--------------|-----|------|-----------|----------------|
| 1       | 72/118       | 72  | 98   | 36.7      | Moderate Risk  |
| 2       | 88/138       | 105 | 95   | 37.2      | High Risk      |
| 3       | 105/170      | 125 | 88   | 39.1      | High Risk      |
| 4       | 65/110       | 68  | 99   | 36.5      | Moderate Risk  |
| 5       | 95/155       | 110 | 91   | 38.4      | High Risk      |

### D. ML-Driven Health Deterioration Prediction System

The Health Deterioration Prediction component was implemented using a Decision Tree classifier trained on ten static demographic and clinical features (Age, Gender, Height, Weight, Smoking, Exercise, Diabetes, Hypertension, Cholesterol, Hereditary predisposition) rather than the sliding-window, trend-based features (moving averages, rate-of-change, SpO2 deviation) specified in the proposed methodology. The binary Low Risk/High Risk target was evaluated on 1,000 held-out samples (142 Low Risk, 858 High Risk). The model achieved a perfect 100% accuracy, precision, recall, and F1-score across both classes (Table V), with Age (50.31%), Smoking status (30.98%), and Cholesterol (17.45%) as dominant features. This requires careful interpretation: perfect accuracy on a clinical task strongly suggests a synthetic or rule-derived dataset, or data leakage, rather than genuine predictive validity. It reflects a static demographic dataset rather than the MIMIC-III trend data originally proposed, so it cannot be compared against proposed target metrics. Despite this, the deployment pipeline serialization, TFLite conversion, and inference was validated end-to-end on synthetic profiles, both correctly classified. Future work should implement the MIMIC-III

pipeline and reevaluate Random Forest against proposed targets.

TABLE V.  
HEALTH DETERIORATION PREDICTION MODEL PERFORMANCE ON TEST SET

| Class        | Precision | Recall | F1-Score | Support |
|--------------|-----------|--------|----------|---------|
| Low Risk     | 1.00      | 1.00   | 1.00     | 142     |
| High Risk    | 1.00      | 1.00   | 1.00     | 858     |
| Accuracy     |           |        | 1.00     | 1000    |
| Macro Avg    | 1.00      | 1.00   | 1.00     | 1000    |
| Weighted Avg | 1.00      | 1.00   | 1.00     | 1000    |

## V. CONCLUSION

Safe Band was developed as an integrated IoT and machine learning based smart wearable ecosystem designed to support emergency health monitoring and response within the Sri Lankan healthcare context. Safe Band incorporated four interconnected components: an ML-Driven Emergency Detection System for real-time critical event classification, an ML-Based Smart Ambulance Dispatch System for nearest hospital identification, an Intelligent AI-Based Hospital Ward Monitoring and Alert Management System for predictive nurse facing risk scoring, and an ML-Driven Health Deterioration Prediction System for advance cardiac risk warning. Experimental evaluation demonstrated that the proposed approaches enabled accurate, low latency, on device detection and decision making across the Emergency Detection, Ambulance Dispatch, and Ward Monitoring components, while also identifying methodological limitations in the Health Deterioration Prediction component that require further refinement. The evaluation results confirm that Safe Band enabled infrastructure light, wearable triggered emergency health monitoring suited to resource constrained healthcare settings. Future work should focus on integrating physiological trend based datasets for the prediction component, conducting end to end latency and usability validation, and testing the system with real clinical users prior to deployment.

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